LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 9 Test (GSEB pattern): Light – Reflection and Refraction

Chapter / Topic: Light – Reflection and Refraction

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SECTION A

Q1. Which of the following can make a parallel beam of light when light from a point source is incident on it? **[1]**

Answer: (a) Concave mirror and convex lens (with the source placed at the focus).

Q2. The radius of curvature of a convex mirror is 32 cm. Its focal length is: **[1]**

Answer: (c) +16 cm. f = R/2, positive for a convex mirror.

Q3. The image formed by a concave mirror is virtual, erect and larger than the object. Where is the object placed? **[1]**

Answer: (d) Between the pole P and F

Q4. Which of the following is used as a magnifying glass? **[1]**

Answer: (d) Convex lens (object placed between F_1 and O)

Q5. The focal length of a lens is +50 cm. Its power is: **[1]**

Answer: (c) +2 D. P = 1/f (in metres) = 1/0.5 m = +2 D

Q6. Light travels fastest in: **[1]**

Answer: (a) vacuum

Q7. The refractive index of glass is 1.5. The speed of light in glass is (speed of light in vacuum = 3×10^8 m/s): **[1]**

Answer: (b) 2×10^8 m/s. v = c/n = $3 \times 10^8/1.5$

Q8. The magnification produced by a plane mirror is: **[1]**

Answer: (b) +1 (image is the same size and erect)

Q9. The mirror used as a rear-view mirror in vehicles is: **[1]**

Answer: (a) convex. It gives an erect image and a wider field of view.

SECTION B

Q10. Define the principal focus of a concave mirror. The radius of curvature of a spherical mirror is 20 cm. What is its focal length? **[2]**

Answer: Rays parallel to the principal axis, after reflection from a concave mirror, all meet at a point on the principal axis called the principal focus. f = R/2 = 20/2 = 10 cm.

- Definition: 1
- f = 10 cm: 1

Q11. Define the power of a lens and give its SI unit. Find the power of a convex lens of focal length 40 cm. **[2]**

Answer: The power of a lens is the reciprocal of its focal length in metres (P = 1/f). It shows how strongly the lens converges or diverges light. SI unit: dioptre (D). P = 1/0.40 m = +2.5 D.

- Definition + unit: 1
- P = +2.5 D: 1

Q12. A ray of light travelling in air enters water at an angle. Does it bend towards or away from the normal? Why? **[2]**

Answer: It bends towards the normal. Light travels more slowly in water, which is optically denser than air.

- Towards the normal: 1
- Reason (slower in denser medium): 1

SECTION C

Q13. An object 4 cm high is placed 15 cm in front of a concave mirror of focal length 10 cm. **[3]**

(i) u = −15 cm, f = −10 cm. 1/v = 1/f − 1/u = −1/10 + 1/15 = −1/30, so v = −30 cm. The image is 30 cm in front of the mirror. [2]

- Formula and correct signs: 1
- v = −30 cm: 1

(ii) m = −v/u = −(−30)/(−15) = −2, so h′ = −2 × 4 = −8 cm. The image is real, inverted and magnified (8 cm tall). [1]

Q14. An object 5 cm high is placed 20 cm in front of a concave mirror of focal length 10 cm. **[3]**

(i) u = −20 cm, f = −10 cm. 1/v = 1/f − 1/u = −1/10 + 1/20 = −1/20, so v = −20 cm. The image is 20 cm in front of the mirror (at C). [2]

- Formula and signs: 1
- v = −20 cm: 1

(ii) m = −v/u = −(−20)/(−20) = −1, so h′ = −5 cm. The image is real, inverted and the same size as the object. [1]

SECTION D

Q15. The refractive index of a medium tells us how much light slows down in it compared with a vacuum. Some values are: water 1.33, kerosene 1.44, crown… **[4]**

(i) Water, which has the lowest refractive index. [1]

(ii) Diamond [1]

(iii) $v = c/n = (3 \times 10^8)/2.42 \approx 1.24 \times 10^8$ m/s [2]

- Formula v = c/n: 1
- Answer with unit: 1